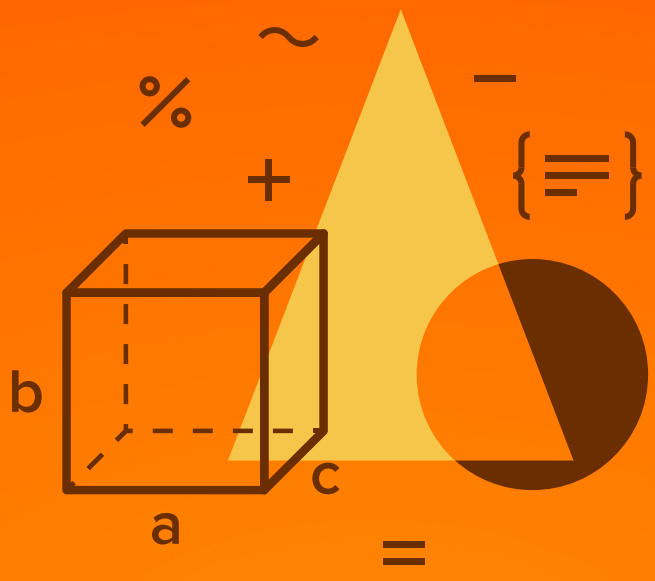
ALLEN CAREER INSTITUTE
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MATHEMATICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Vector-3D

English Medium

EXERCISE (O-1)

Straight Objective Type

1. The intercept made by the plane $\vec{r}.\vec{n}=q$ on the x-axis is

(A) $\frac{q}{\hat{i}.\vec{n}}$ (B) $\frac{\hat{i}.\vec{n}}{q}$ (C) $\left(\hat{i}.\vec{n}\right)q$ (D) $\frac{q}{|\vec{n}|}$

TD0002

2. If the distance between the planes

$8x + 12y – 14z = 2$

and $4x + 6y – 7z = 2$

can be expressed in the form $\frac{1}{\sqrt{N}}$ where N is natural then the value of $\frac{N(N+1)}{2}$ is

(A) 4950 (B) 5050 (C) 5150 (D) 5151

TD0003

3. A plane passes through the point P(4, 0, 0) and Q(0, 0, 4) and is parallel to the y-axis. The distance of the plane from the origin is

(A) 2 (B) 4 (C) $\sqrt{2}$ (D) $2\sqrt{2}$

TD0004

4. If from the point P (f, g, h) perpendiculars PL, PM be drawn to yz and zx planes then the equation to the plane OLM is

(A) $\frac{x}{f}+\frac{y}{g}-\frac{z}{h}=0$ (B) $\frac{x}{f}+\frac{y}{g}+\frac{z}{h}=0$ (C) $\frac{x}{f}-\frac{y}{g}+\frac{z}{h}=0$ (D) $-\frac{x}{f}+\frac{y}{g}+\frac{z}{h}=0$

TD0005

5. If the plane $2x – 3y + 6z – 11 = 0$ makes an angle $\sin^{-1}(k)$ with x-axis, then k is equal to

(A) $\sqrt{3}/2$ (B) 2/7 (C) $\sqrt{2}/3$ (D) 1

TD0006

6. The plane XOZ divides the join of (1, –1, 5) and (2, 3, 4) in the ratio $\lambda : 1$, then λ is

(A) – 3 (B) – 1/3 (C) 3 (D) 1/3

TD0007

7. A variable plane forms a tetrahedron of constant volume 64 K^3 with the coordinate planes and the origin, then locus of the centroid of the tetrahedron is

(A) $x^3 + y^3 + z^3 = \pm 6K^3$ (B) $xyz = \pm 6K^3$

(C) $x^2 + y^2 + z^2 = 4K^2$ (D) $x^{-2} + y^{-2} + z^{-2} = 4K^{-2}$

TD0008

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8. Let ABCD be a tetrahedron such that the edges AB, AC and AD are mutually perpendicular. Let the area of triangles ABC, ACD and ADB be 3, 4 and 5 sq. units respectively. Then the area of the triangle BCD, is

(A) $5\sqrt{2}$ (B) 5 (C) $5/\sqrt{2}$ (D) 5/2

TD0009

9. Equation of the line which passes through the point with p. v. (2, 1, 0) and perpendicular to the plane containing the vectors $\hat{i}+\hat{j}$ and $\hat{j}+\hat{k}$ is

(A) $\vec{r}$ = (2, 1, 0) + t (1, –1, 1) (B) $\vec{r}$ = (2, 1, 0) + t (–1, 1, 1)

(C) $\vec{r}$ = (2, 1, 0) + t (1, 1, –1) (D) $\vec{r}$ = (2, 1, 0) + t (1, 1, 1)

where t is a parameter

TD0010

10. Which of the following planes are parallel but not identical?

$P_1 : 4x - 2y + 6z = 3$

$P_2 : 4x - 2y - 2z = 6$

$P_3 : -6x + 3y - 9z = 5$

$P_4 : 2x - y - z = 3$

(A) P_2 & P_3 (B) P_2 & P_4 (C) P_1 & P_3 (D) P_1 & P_4

TD0011

11. A parallelopiped is formed by planes drawn through the points (1, 2, 3) and (9, 8, 5) parallel to the coordinate planes then which of the following is not the length of an edge of this rectangular parallelopiped

(A) 2 (B) 4 (C) 6 (D) 8

TD0012

12. Vector equation of the plane $\vec{r}=\hat{i}-\hat{j}+\lambda(\hat{i}+\hat{j}+\hat{k})+\mu(\hat{i}-2\hat{j}+3\hat{k})$ in the scalar dot product form is

(A) $\vec{r}.(5\hat{i}-2\hat{j}+3\hat{k})=7$ (B) $\vec{r}.(5\hat{i}+2\hat{j}-3\hat{k})=7$

(C) $\vec{r}.(5\hat{i}-2\hat{j}-3\hat{k})=7$ (D) $\vec{r}.(5\hat{i}+2\hat{j}+3\hat{k})=7$

TD0013

13. The vector equations of the two lines L_1 and L_2 are given by

$L_1 : \vec{r} = 2\hat{i}+9\hat{j}+13\hat{k}+\lambda(\hat{i}+2\hat{j}+3\hat{k})$; $L_2 : \vec{r} = -3\hat{i}+7\hat{j}+p\hat{k}+\mu(-\hat{i}+2\hat{j}-3\hat{k})$

then the lines L_1 and L_2 are

(A) skew lines for all $p \in R$

(B) intersecting for all $p \in R$ and the point of intersection is (–1, 3, 4)

(C) intersecting lines for p = – 2

(D) intersecting for all real $p \in R$

TD0014

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14. Consider the plane $(x, y, z) = (0, 1, 1) + \lambda(1, -1, 1) + \mu(2, -1, 0)$. The distance of this plane from the origin is

(A) 1/3 (B) $\sqrt{3}/2$ (C) $\sqrt{3/2}$ (D) $2/\sqrt{3}$

TD0015

15. The value of 'a' for which the lines $\frac{x-2}{1}=\frac{y-9}{2}=\frac{z-13}{3}$ and $\frac{x-a}{-1}=\frac{y-7}{2}=\frac{z+2}{-3}$ intersect, is

(A) –5 (B) –2 (C) 5 (D) –3

TD0016

16. Given A (1, –1, 0) ; B(3, 1, 2) ; C(2, –2, 4) and D(–1, 1, –1) which of the following points neither lie on AB nor on CD?

(A) (2, 2, 4) (B) (2, –2, 4) (C) (2, 0,1) (D) (0, –2, –1)

TD0017

17. For the line $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$, which one of the following is incorrect?

(A) it lies in the plane x – 2y + z = 0 (B) it is same as line $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$

(C) it passes through (2, 3, 5) (D) it is parallel to the plane x – 2y + z – 6 = 0

TD0018

18. Given planes $P_1 : cy + bz = x$

$P_2 : az + cx = y$

$P_3 : bx + ay = z$

P_1, P_2 and P_3 pass through one line, if

(A) $a^2 + b^2 + c^2 = ab + bc + ca$ (B) $a^2 + b^2 + c^2 + 2abc = 1$

(C) $a^2 + b^2 + c^2 = 1$ (D) $a^2 + b^2 + c^2 + 2ab + 2bc + 2ca + 2abc = 1$

TD0019

19. The line which contains all points (x, y, z) which are of the form $(x, y, z) = (2, -2, 5) + \lambda(1, -3, 2)$ intersects the plane 2x – 3y + 4z = 163 at P and intersects the YZ plane at Q. If the distance PQ is $a\sqrt{b}$ where $a, b \in N$ and $a > 3$ then (a + b) equals

(A) 23 (B) 95 (C) 27 (D) none

TD0022

20. Let L_1 be the line $\vec{r}_1 = 2\hat{i} + \hat{j} - \hat{k} + \lambda(\hat{i} + 2\hat{k})$ and let L_2 be the line $\vec{r}_2 = 3\hat{i} + \hat{j} + \mu(\hat{i} + \hat{j} - \hat{k})$.

Let Π be the plane which contains the line L_1 and is parallel to L_2. The distance of the plane Π from the origin is

(A) 1/7 (B) $\sqrt{2/7}$ (C) $\sqrt{6}$ (D) none

TD0023

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21. The value of m for which straight line $3x - 2y + z + 3 = 0 = 4x - 3y + 4z + 1$ is parallel to the plane $2x - y + mz - 2 = 0$ is

(A) –2 (B) 8 (C) – 18 (D) 11

TD0024

22. The distance of the point (–1, –5, – 10) from the point of intersection of the line $\frac{x-2}{2}=\frac{y+1}{4}=\frac{z-2}{12}$ and the plane $x - y + z = 5$ is

(A) $2\sqrt{11}$ (B) $\sqrt{126}$ (C) 13 (D) 14

TD0025

23. $P(\vec{p})$ and $Q(\vec{q})$ are the position vectors of two fixed points and $R(\vec{r})$ is the position vector of a variable point. If R moves such that $(\vec{r}-\vec{p})\times(\vec{r}-\vec{q})=\vec{0}$ then the locus of R is

(A) a plane containing the origin 'O' and parallel to two non collinear vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$

(B) the surface of a sphere described on PQ as its diameter.

(C) a line passing through the points P and Q

(D) a set of lines parallel to the line PQ. **TD0026**

24. Let the distance between the plane $3x + 4y + 5z + c_1 = 0$ and $3x + 4y + 5z - c_2 = 0$ is $2\sqrt{2}$, where $c_1, c_2 \in W$, then number of possible ordered pair(s) of (c_1, c_2) is - **TD0139**

(A) 1 (B) 21 (C) 31 (D) 51

25. Consider the lines $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ and $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$, then the equation of the line which

(A) bisects the angle between the lines is $\frac{x}{3}=\frac{y}{3}=\frac{z}{8}$

(B) bisects the angle between the lines is $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$

(C) passes through origin and is perpendicular to the given lines is $x = y = -z$

(D) none of these **TD0153**

26. A plane meets the coordinate axes in A, B, C and (α, β, γ) is the centroid of the triangle ABC, then the equation of the plane is

(A) $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=3$ (B) $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=1$

(C) $\frac{3x}{\alpha}+\frac{3y}{\beta}+\frac{3z}{\gamma}=1$ (D) $\alpha x + \beta y + \gamma z = 1$ **TD0154**

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27. Equation of plane which passes through the point of intersection of lines $\frac{x-1}{3}=\frac{y-2}{1}=\frac{z-3}{2}$ and $\frac{x-3}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ and at greatest distance from the point (0, 0, 0) is:

(A) $4x + 3y + 5z = 25$ (B) $4x + 3y + 5z = 50$

(C) $3x + 4y + 5z = 49$ (D) $x + 7y - 5z = 2$ **TD0155**

28. The non zero value of 'a' for which the lines $2x - y + 3z + 4 = 0 = ax + y - z + 2$ and $x - 3y + z = 0 = x + 2y + z + 1$ are co-planar is :

(A) –2 (B) 4 (C) 6 (D) 0

TD0156

29. A line having direction ratios 3, 4, 5 cuts 2 planes $2x-3y+6z-12=0$ and $2x-3y+6z+2=0$ at point P & Q, then find length of PQ

(A) $\frac{35\sqrt{2}}{12}$ (B) $\frac{35\sqrt{2}}{24}$ (C) $\frac{35\sqrt{2}}{6}$ (D) $\frac{35\sqrt{2}}{8}$

TD0157

30. A line L_1 having dierection ratios 1, 0, 1 lies on xz plane. Now this xz plane is rotated about z-axis by an angle of 90°. Now the new position of L_1 is L_2. The angle between L_1 & L_2 is :

(A) 30° (B) 60° (C) 90° (D) 45°

TD0158

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EXERCISE (O-2)

Multiple Correct Answer Type

1. Equation of plane containing the line $2x-y+z-3=0=3x+y+z-5$ and lying at a distance $\frac{1}{\sqrt{6}}$ units from the point (2,1,–1) can be- **TD0141**

(A) $2x-y+z-3=0$ (B) $62x+29y+19z-105=0$

(C) $2x+y-z-5=0$ (D) $62x+29y-19z+104=0$

2. The volume of a right triangular prism $ABCA_1B_1C_1$ is equal to 3. If the position vectors of the vertices of the base ABC are A(1, 0, 1) ; B(2,0, 0) and C(0, 1, 0) then the position vectors of the vertex A_1 can be:

(A) (2, 2, 2) (B) (0, 2, 0) (C) (0, –2, 2) (D) (0, - 2, 0)

TD0031

3. Consider Lines $L_1:\frac{x-\alpha}{1}=\frac{y}{-2}=\frac{z+\beta}{2}$, $L_2: x=\alpha, \frac{y}{-\alpha}=\frac{z+\alpha}{2-\beta}$, plane $P: 2x+2y+z+7=0$. Let line L_2 lies in plane P, then

(A) $\alpha=-7$

(B) $\alpha=7$

(C) minimum distance between line L_1 and plane P is 11.

(D) minimum distance between line L_1 and plane P is $\frac{23}{3}$

TD0032

4. The point $\vec{A}(3,4,7)$, $\vec{B}(4,5,9)$ and $\vec{C}(1,2,-1)$ are three vertices of a parallelogram ABCD, then -

(A) vector equation of line AB is $\vec{r}=4\hat{i}+5\hat{j}+9\hat{k}+\lambda\left(\hat{i}+\hat{j}+2\hat{k}\right)$

(B) cartesian equation of line BC is $\frac{x-4}{3}=\frac{y-5}{3}=\frac{z-9}{10}$

(C) coordinates of D are (0,1,–3)

(D) ABCD is rectangle

TD0033

5. Let two planes $P_1: 2x-y+z-2=0$ and $P_2: x+2y-z-3=0$ are given then-

(A) The equation of the plane through line of intersection of $P_1=0$ and $P_2=0$ and the point (3,2,1) is $x-3y+2z+1=0$

(B) The equation of the plane through line of intersection of $P_1=0$ and $P_2=0$ and the point (3,2,1) is $3x-y+2z-9=0$

(C) The equation of acute angle bisector plane of $P_1=0$ and $P_2=0$ is $x-3y+2z+1=0$

(D) The equation of acute angle bisector plane of $P_1=0$ and $P_2=0$ is $x+3y+2z+2=0$

TD0034

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6. A variable point P(4 cost, 4sint, 4sint) moves in space, now which of the following holds good ?

(A) Point P moves on a plane

(B) Point 'P' traces a circle.

(C) Area enclosed by P is $16\sqrt{2}\pi$

(D) Point P cannot lie on a fixed plane.

TD0035

7. The projection of line $\frac{x}{2}=\frac{y-1}{2}=\frac{z-1}{1}$ on a plane 'P' is $\frac{x}{1}=\frac{y-1}{1}=\frac{z-1}{-1}$. If the plane P passes through (k, –2, 0), then k is greater than -

(A) 2 (B) 3 (C) 5 (D) 4

TD0036

8. A line segment has length 6 and direction ratios are –3,4,6, then the component of the line vector are-

(A) $\frac{-18}{\sqrt{61}},\frac{24}{\sqrt{61}},\frac{36}{\sqrt{61}}$

(B) 27,–18,–54

(C) 27,–18,54

(D) $\frac{18}{\sqrt{61}},\frac{-24}{\sqrt{61}},\frac{-36}{\sqrt{61}}$

TD0037

9. Which of the following is (are) correct -

(A) If two lines in space are not intersecting, then they must be skew lines.

(B) If two lines are parallel to a plane 'P', then their direction ratios will be proportional

(C) If two lines are perpendicular to a plane 'P', then their direction ratios will be proportional

(D) Equation $\frac{x+1}{a}=\frac{y-1}{b}=\frac{z}{c}$, where a, b, c are real parameters, represents a family of concurrent lines in space

TD0038

10. Given the equations of the line $3x - y + z + 1 = 0,\ 5x + y + 3z = 0$.

Then which of the following is correct ?

(A) Symmetrical form of the equations of line is $\frac{x}{2}=\frac{y-\frac{1}{8}}{-1}=\frac{z+\frac{5}{8}}{1}$

(B) symmetrical form of the equations of line is $\frac{x+\frac{1}{8}}{1}=\frac{y-\frac{5}{8}}{1}=\frac{z}{-2}$

(C) equation of the plane through (2, 1, 4) and prependicular to the given lines is $2x - y + z - 7 = 0$

(D) equation of the plane through (2, 1, 4) and prependicular to the given lines is $x + y - 2z + 5 = 0$

TD0039

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11. Consider the family of planes $x + y + z = c$ where is a parameter intersecting the coordinate axes at P, Q,R and α,β,γ are the angles made by each member of this family with positive x,y and z axis. Which of the following interpretations hold good for this family -

(A) each member of this family is equally inclined with the coordinate axes.

(B) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$

(C) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 2$

(D) for $c = 3$ area of the triangle PQR is $3\sqrt{3}$ sq. units.

TD0040

12. Consider a plane P passing through $A(\lambda,3,\mu)$, $B(-1,3,2)$ and $C(7,5,10)$ and a straight line L with positive direction cosines passing through A, bisecting BC and makes equal angles with the coordinate axes. Let L_1 be a line parallel to L and passing through origin. Which of the following is(are) correct?

(A) The value of $(\lambda + \mu)$ is equal to 5.

(B) Equation of straight line L_1 is $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}$.

(C) Equation of the plane perpendicular to the plane P and containing line L_1 is $x - 2y + z = 0$

(D) Area of triangle ABC is equal to $3\sqrt{2}$.

TD0041

13. A line L passing through the point P(1,4,3), is perpendicular to both the lines

$\frac{x-1}{2} = \frac{y+3}{1} = \frac{z-2}{4}$ and $\frac{x+2}{3} = \frac{y-4}{2} = \frac{z+1}{-2}$.

If the position vector of point Q on L is (a_1,a_2,a_3) such that $(PQ)^2 = 357$, then $(a_1 + a_2 + a_3)$ can be-

(A) 16 (B) 15 (C) 2 (D) 1

TD0042

14. Which of the following is (are) correct -

(A) If two lines in space are not intersecting, then they must be skew lines.

(B) If two lines are parallel to a plane 'P', then their direction ratios will be proportional

(C) If two lines are perpendicular to a plane 'P', then their direction ratios will be proportional

(D) Equation $\frac{x+1}{a} = \frac{y-1}{b} = \frac{z}{c}$, where a, b, c are real parameters, repres ents a family of concurrent lines in space

TD0142

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15. The direction cosines of the lines bisecting the angle between the lines whose direction cosines are ℓ_1, m_1, n_1 and ℓ_2, m_2, n_2 and the angle between these lines is θ, are

(A) $\frac{\ell_1-\ell_2}{\cos\frac{\theta}{2}}, \frac{m_1-m_2}{\cos\frac{\theta}{2}}, \frac{n_1-n_2}{\cos\frac{\theta}{2}}$

(B) $\frac{\ell_1+\ell_2}{2\cos\frac{\theta}{2}}, \frac{m_1+m_2}{2\cos\frac{\theta}{2}}, \frac{n_1+n_2}{2\cos\frac{\theta}{2}}$

(C) $\frac{\ell_1+\ell_2}{2\sin\frac{\theta}{2}}, \frac{m_1+m_2}{2\sin\frac{\theta}{2}}, \frac{n_1+n_2}{2\sin\frac{\theta}{2}}$

(D) $\frac{\ell_1-\ell_2}{2\sin\frac{\theta}{2}}, \frac{m_1-m_2}{2\sin\frac{\theta}{2}}, \frac{n_1-n_2}{2\sin\frac{\theta}{2}}$ **TD0159**

16. The vertices of a triangle are A(1, 1, 2), B(4, 3, 1) and C(2, 3, 5). A vector representing the bisector of the angle A is :

(A) $2\hat{i}-4\hat{k}$ (B) $-2\hat{i}+4\hat{k}$ (C) $-2\hat{i}-2\hat{j}-\hat{k}$ (D) $2\hat{i}+2\hat{j}+\hat{k}$

TD0160

17. Acute angle between the lines $\frac{x-1}{\ell}=\frac{y+1}{m}=\frac{z}{n}$ and $\frac{x+1}{m}=\frac{y-3}{n}=\frac{z-1}{\ell}$ where $\ell > m > n$, and ℓ, m, n are the roots of the cubic equation $x^3+x^2-4x=4$ is equal to :

(A) $\cos^{-1}\frac{3}{\sqrt{13}}$ (B) $\sin^{-1}\frac{\sqrt{65}}{9}$ (C) $2\cos^{-1}\sqrt{\frac{13}{18}}$ (D) $\tan^{-1}\frac{2}{3}$

TD0161

18. The line $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-1}{-1}$ intersects the curve $xy=c^2, z=0$ if c is equal to :

(A) -1 (B) $-\sqrt{5}$ (C) $\sqrt{5}$ (D) 1

TD0162

19. Three distinct lines $\frac{x-1}{3}=\frac{y-2}{2}=\frac{z-3}{1}$, $\frac{x-1}{5}=\frac{2y-4}{3}=\frac{3z-9}{1}$, $\frac{x-\lambda^2}{3}=\frac{y-2}{2}=\frac{z-3}{\lambda}$ are concurrent the value of λ may be :

(A) 1 (B) -1 (C) 2 (D) -2

TD0163

20. The line $\frac{x+6}{5}=\frac{y+10}{3}=\frac{z+14}{8}$ is the hypotenuse of an isosceles right angle triangle whose opposite vertex is (7, 2, 4) Then the equation of remaining sides is/are -

(A) $\frac{x-7}{3}=\frac{y-2}{6}=\frac{z-4}{2}$

(B) $\frac{x-7}{2}=\frac{y-2}{-3}=\frac{z-4}{6}$

(C) $\frac{x+7}{3}=\frac{y+2}{6}=\frac{z+4}{2}$

(D) $\frac{x+7}{2}=\frac{y+2}{-3}=\frac{z+4}{6}$ **TD0164**

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EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 and 2

Consider planes $P_1 : x + 2y + z - 1 = 0$ & $P_2 : x - y + 2z + 1 = 0$ and a plane P containing the line of intersection of P_1 & P_2.

1. If P is perpendicular to the plane $x + y + z = 5$, then equation of P is -

(A) $2x - 4y + 3z = 3$ (B) $x + 4y + z = 3$

(C) $x - 4y + 3z + 3 = 0$ (D) $-x - 4y + 4z + 3 = 0$ **TD0143**

2. If from a point Q lying on plane P in I[st] octant the distances of planes $x = 0$, $y = 0$ & $z = 0$ are 1, 1 and 2 respectively, then vector equation of line passing through origin and parallel to normal of the plane P is -

(A) $\vec{r} = \lambda(\hat{i} - 14\hat{j} + 2\hat{k})$ (B) $\vec{r} = \lambda(2\hat{i} - \hat{j} + \hat{k})$

(C) $\vec{r} = \lambda(\hat{i} - \hat{j} + \hat{k})$ (D) $\vec{r} = \lambda(\hat{i} + 14\hat{j} - 3\hat{k})$ **TD0144**

(where $\lambda \in R$)

Paragraph for Question 3 & 4

Consider a plane $\Pi : \vec{r}.(\hat{i} - \hat{j} + 2\hat{k}) = 5$, a line $L_1 : \vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + \lambda(4\hat{i} + \hat{j} - \hat{k})$ and a point A(2,2,1). L_2 is a line passing through A intersecting L_1 and parallel to plane Π -

3. Equation of L_2 is - **TD0145**

(A) $\vec{r} = (2 + 6\lambda)\hat{i} - 2\hat{j} + \lambda\hat{k}$ (B) $\vec{r} = 2\hat{i} - (2 + 6\lambda)\hat{j} + \hat{k}$

(C) $\vec{r} = (4\lambda + 6)\hat{i} + 2\lambda\hat{j} - \lambda\hat{k}$ (D) $\vec{r} = (2 + 6\lambda)\hat{i} + (\lambda - 3)\hat{j} + (2 + \lambda)\hat{k}$

4. Line L_1 intersects plane Π at Q and xy-plane at R, then volume of tetrahedron OAQR is (where O is origin) **TD0146**

(A) 1 cubic units (B) $\frac{1}{2}$ cubic units (C) 6 cubic units (D) 4 cubic units

Paragraph for Question 5 & 6

A conic section is obtained by the intersection of two inverted cones (having same vertex) with a plane. If a plane passes through the vertex, then its intersection with the cones either represents a pair of straight lines or a point depending upon whether it intersects the cones or not. If plane does not pass through vertex, then section may be hyperbola or circle depending upon whether plane is parallel to axis of the cone or perpendicular to it. If it has any other inclination, then intersection of plane and cone gives an ellipse or a parabola.

5. If a variable point P moves such that the line passing through P and Q (0,0,2) makes an angle θ with z-axis, then the locus of P is :

(A) z-axis if $\theta = 0°$ (B) Plane $z = 2$ if $\theta = 90°$

(C) A cone $x^2 + y^2 - 3(z - 2)^2 = 0$ if $\theta = 60°$ (D) A cone $x^2 + y^2 + 3(z - 2)^2 = 0$ if $\theta = 60°$

TD0165

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6. In the above question if $\theta = 60°$ then the locus of intersection of locus of P and $ax + y + z = b$ will be

(A) a hyperbola if $a = b = 1$ (B) a pair of straight line if $a = 1$ and $b = 2$

(C) a pair of straight lines if $a = b = 2$ (D) none of these **TD0166**

Paragraph for Question 7 & 8

In 3D geometry, a tetrahedron is a four-faced polyhedron with four vertices. It is one of the simplest and most important shapes in 3D geometry, and it has many practical applications in fields such as engineering, physics, chemistry and computer graphics. To solve problems related to a tetrahedron, we need to use concepts such as shortest distance formula, cross product, and the formula for the volume of a tetrahedron. By applying these concepts, we can gain a better understanding of 3D geometry and solve problems related to the tetrahedron.

7. Find the shortest distance between any two opposite edges of a tetrahedron formed by the planes $y + z = 0,\ x + z = 0,\ x + y = 0,\ x + y + z = \sqrt{3}$. **TD0167**

8. If the volume of tetrahedron formed by planes whose equations are $y + z = 0$, $z + x = 0$, $x + y = 0$ and $x + y + z = 1$ is t cubic unit, then the value of t is **TD0168**

Matrix Match Type

9. Consider the following four pairs of lines in **List-I** and match them with one or more entries in **List-II**.

	List-I		**List-II**
(I)	$L_1 : x = 1 + t,\ y = t,\ z = 2 - 5t$ $L_2 : \vec{r} = (2,1,-3) + \lambda(2, 2, -10)$	(P)	non coplanar lines
(II)	$L_1 : \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$ $L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z+2}{3}$	(Q)	lines lie in a unique plane
(III)	$L_1 : x = -6t,\ y = 1 + 9t,\ z = -3t$ $L_2 : x = 1 + 2s,\ y = 4 - 3s,\ z = s$	(R)	infinite planes containing both the lines
(IV)	$L_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ $L_2 : \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$	(S)	lines are not intersecting at a unique point

TD0027 **TD0028** **TD0029** **TD0030**

(A) I $\to$ Q,S; II $\to$ P,S; III $\to$ R,S; IV $\to$ Q (B) I $\to$ Q; II $\to$ Q,S; III $\to$ P,S; IV $\to$ R,S

(C) I $\to$ P,S; II $\to$ Q,S; III $\to$ Q; IV $\to$ R,S (D) I $\to$ R,S; II $\to$ Q; III $\to$ Q,S; IV $\to$ P,S

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10. P(0, 3, –2); Q(3, 7, –1) and R(1, –3, –1) are 3 given points. Let L_1 be the line passing through P and Q and L_2 be the line through R and parallel to the vector $\vec{V} = \hat{i} + \hat{k}$.

	Column-I		**Column-II**
(A)	perpendicular distance of P from L_2	(P)	$7\sqrt{3}$
(B)	shortest distance between L_1 and L_2	(Q)	2
(C)	area of the triangle PQR	(R)	6
(D)	distance from (0, 0, 0) to the plane PQR	(S)	$\frac{19}{\sqrt{147}}$

TD0043

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EXERCISE (O-4)

Numerical Grid Type

1. For what value of p the lines $\frac{x+1}{-3}=\frac{y-p}{2}=\frac{z+2}{1}$ and $\frac{x}{1}=\frac{y-7}{-3}=\frac{z+7}{2}$ are in the same plane.

TD0048

2. The equation of the plane which has the property that the point Q(5,4,5) is the reflection of point P(1,2,3) through that plane, is ax + by + cz = d where a,b,c,d $\in$ N. Find the least value of (a + b + c + d).

TD0056

3. A line from origin meets the lines $\frac{x-1}{1}=\frac{y}{2}=\frac{z+1}{2}$ and $\frac{x}{2}=\frac{y-1}{3}=\frac{z}{1}$ at A and B, then length of AB is equal to **TD0169**

4. Let the equation of locus of a point which moves such that its distance from (1,1,1) is double its distance from x axis is $x^2-\lambda_1y^2-\lambda_2z^2-2x-2y-2z+\lambda_3=0$, then $|\lambda_1|+|\lambda_2|+|\lambda_3|$ is equal to

TD0170

5. If the distance between the plane Ax – 2y + z = d and the plane containing the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ is $\sqrt{6}$, then |d| is **TD0105**

6. If the distance between the plane, 23x – 10y – 2z + 48 = 0 and the plane containing the lines $\frac{x+1}{2}=\frac{y-3}{4}=\frac{z+1}{3}$ and $\frac{x+3}{2}=\frac{y+2}{6}=\frac{z-1}{\lambda}(\lambda\in R)$ is equal to $\frac{k}{\sqrt{633}}$, then k is equal to _____.

TD0101

7. If the lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-1}{4}$ and $\frac{x-3}{1}=\frac{y-k}{2}=\frac{z}{1}$ intersect, then k is equal to :

TD0074

8. Distance between two parallel planes 2x + y + 2z = 8 and 4x + 2y + 4z + 5 = 0 is :-

TD0075

9. If the foot of the perpendicular drawn from the point (1, 0, 3) on a line passing through (α, 7, 1) is $\left(\frac{5}{3},\frac{7}{3},\frac{17}{3}\right)$, then α is equal to_____

TD0097

10. A line from the origin meets the lines $\frac{x-2}{1}=\frac{y-1}{-2}=\frac{z+1}{1}$ and $\frac{x-\frac{8}{3}}{2}=\frac{y+3}{-1}=\frac{z-1}{1}$ at P and Q respectively. If length PQ = d, then d is **TD0106**

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EXERCISE (JM)

1. An equation of a plane parallel to the plane $x - 2y + 2z - 5 = 0$ and at a unit distance from the origin is : **[AIEEE-2012]**

(1) $x - 2y + 2z + 5 = 0$ (2) $x - 2y + 2z - 3 = 0$

(3) $x - 2y + 2z + 1 = 0$ (4) $x - 2y + 2z - 1 = 0$

TD0073

2. Let Q be the foot of perpendicular from the origin to the plane $4x - 3y + z + 13 = 0$ and R be a point $(-1, 1, -6)$ on the plane. Then length QR is :- **[JEE-MAIN Online 2013]**

(1) $3\sqrt{\frac{7}{2}}$ (2) $\sqrt{14}$ (3) $\sqrt{\frac{19}{2}}$ (4) $\frac{3}{\sqrt{2}}$

TD0079

3. If the projections of a line segment on thex, y and z-axes in 3-dimensional space are 2, 3 and 6 respectively, then the length ofthe line segment is : **[JEE-MAIN Online 2013]**

(1) 7 (2) 9 (3) 12 (4) 6

TD0080

4. The equation of a plane through the line of intersection of the planes $x + 2y = 3$, $y - 2z + 1 = 0$, and perpendicular to the first plane is : **[JEE-MAIN Online 2013]**

(1) $2x - y + 7z = 11$ (2) $2x - y + 10z = 11$ (3) $2x - y - 9z = 10$ (4) $2x - y - 10z = 9$

TD0082

5. Let ABC be a triangle with vertices at points A (2, 3, 5), B (-1, 3, 2) and C $(\lambda, 5, \mu)$ in three dimensional space. If the median through A is equally inclined with the axes, then $(\lambda, \mu.)$ is equal to : **[JEE-MAIN Online 2013]**

(1) (10, 7) (2) (7.5) (3) (7, 10) (4) (5,7)

TD0083

6. The angle between the lines whose direction cosines satisfy the equations $\ell + m + n = 0$ and $\ell^2 = m^2 + n^2$ is : **[JEE-MAIN 2014]**

(1) $\frac{\pi}{3}$ (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{2}$

TD0084

7. The image of the line $\frac{x-1}{3} = \frac{y-3}{1} = \frac{z-4}{-5}$ in the plane $2x - y + z + 3 = 0$ is the line : **[JEE-MAIN 2014]**

(1) $\frac{x+3}{3} = \frac{y-5}{1} = \frac{z-2}{-5}$ (2) $\frac{x+3}{-3} = \frac{y-5}{-1} = \frac{z+2}{5}$

(3) $\frac{x-3}{3} = \frac{y+5}{1} = \frac{z-2}{-5}$ (4) $\frac{x-3}{-3} = \frac{y+5}{-1} = \frac{z-2}{5}$

TD0085

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8. The equation of the plane containing the line $2x - 5y + z = 3$; $x + y + 4z = 5$, and parallel to the plane, $x + 3y + 6z = 1$, is : **[JEE(Main)-2015]**

(1) $x + 3y + 6z = 7$ (2) $2x + 6y + 12z = -13$

(3) $2x + 6y + 12z = 13$ (4) $x + 3y + 6z = -7$

TD0086

9. The distance of the point (1, 0, 2) from the point of intersection of the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane $x - y + z = 16$, is : **[JEE(Main)-2015]**

(1) $3\sqrt{21}$ (2) 13 (3) $2\sqrt{14}$ (4) 8

TD0087

10. The distance of the point (1, –5, 9) from the plane $x - y + z = 5$ measured along the line $x = y = z$ is : **[JEE(Main)-2016]**

(1) $\frac{20}{3}$ (2) $3\sqrt{10}$ (3) $10\sqrt{3}$ (4) $\frac{10}{\sqrt{3}}$

TD0088

11. If the image of the point P(1, –2, 3) in the plane, $2x + 3y - 4z + 22 = 0$ measured parallel to line, $\frac{x}{1} = \frac{y}{4} = \frac{z}{5}$ is Q, then PQ is equal to :- **[JEE(Main)-2017]**

(1) $6\sqrt{5}$ (2) $3\sqrt{5}$ (3) $2\sqrt{42}$ (4) $\sqrt{42}$

TD0089

12. The distantce of the point (1, 3, –7) from the plane passing through the point (1, –1, –1), having normal perpendicular to both the lines $\frac{x-1}{1} = \frac{y+2}{-2} = \frac{z-4}{3}$ and $\frac{x-2}{2} = \frac{y+1}{-1} = \frac{z+7}{-1}$, is :- **[JEE(Main)-2017]**

(1) $\frac{10}{\sqrt{74}}$ (2) $\frac{20}{\sqrt{74}}$ (3) $\frac{10}{\sqrt{83}}$ (4) $\frac{5}{\sqrt{83}}$

TD0090

13. The length of the projection of the line segment joining the points (5, –1, 4) and (4, –1, 3) on the plane, $x + y + z = 7$ is : **[JEE(Main)-2018]**

(1) $\frac{2}{3}$ (2) $\frac{1}{3}$ (3) $\sqrt{\frac{2}{3}}$ (4) $\frac{2}{\sqrt{3}}$

TD0091

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14. If L_1 is the line of intersection of the planes $2x - 2y + 3z - 2 = 0$, $x - y + z + 1 = 0$ and L_2 is the line of intersection of the planes $x + 2y - z - 3 = 0$, $3x - y + 2z - 1 = 0$, then the distance of the origin from the plane, containing the lines L_1 and L_2 is : **[JEE(Main)-2018]**

(1) $\frac{1}{3\sqrt{2}}$ (2) $\frac{1}{2\sqrt{2}}$ (3) $\frac{1}{\sqrt{2}}$ (4) $\frac{1}{4\sqrt{2}}$

TD0092

15. The equation of the line passing through $(-4, 3, 1)$, parallel to the plane $x + 2y - z - 5 = 0$ and intersecting the line $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z-2}{-1}$ is : **[JEE(Main)-19]**

(1) $\frac{x+4}{-1} = \frac{y-3}{1} = \frac{z-1}{1}$ (2) $\frac{x+4}{3} = \frac{y-3}{-1} = \frac{z-1}{1}$ (3) $\frac{x+4}{1} = \frac{y-3}{1} = \frac{z-1}{3}$ (4) $\frac{x-4}{2} = \frac{y+3}{1} = \frac{z+1}{4}$

TD0093

16. If the lines $x = ay + b$, $z = cy + d$ and $x = a'z + b'$, $y = c'z + d'$ are perpendicular, then : **[JEE(Main)-19]**

(1) $cc' + a + a' = 0$ (2) $aa' + c + c' = 0$ (3) $ab' + bc' + 1 = 0$ (4) $bb' + cc' + 1 = 0$

TD0094

17. A plane which bisects the angle between the two given planes $2x - y + 2z - 4 = 0$ and $x + 2y + 2z - 2 = 0$, passes through the point : **[JEE(Main)- 19]**

(1) (2,4,1) (2) (2, – 4, 1) (3) (1, 4, – 1) (4) (1, – 4, 1)

TD0096

18. Let P be a plane passing through the points (2, 1, 0), (4, 1, 1) and (5, 0, 1) and R be any point (2, 1, 6). Then the image of R in the plane P is : **[JEE(Main)- 20]**

(1) (6, 5, –2) (2) (4, 3, 2) (3) (3, 4, –2) (4) (6, 5, 2)

TD0098

19. The mirror image of the point (1,2,3) in a plane is $\left(-\frac{7}{3}, -\frac{4}{3}, -\frac{1}{3}\right)$. Which of the following points lies on this plane ? **[JEE(Main)- 20]**

(1) (–1, –1, –1) (2) (–1, –1, 1) (3) (1, 1, 1) (4) (1, –1, 1)

TD0099

20. The shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is **[JEE(Main)- 20]**

(1) $\frac{7}{2}\sqrt{30}$ (2) $3\sqrt{30}$ (3) 3 (4) $2\sqrt{30}$

TD0100

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Maths\Module\Vol-Module\#Matrices, Vector & 3D\Eng\08-3D_Exe.p65

21. If for some α and β in R, the intersection of the following three places
$x + 4y - 2z = 1$
$x + 7y - 5z = \beta$
$x + 5y + \alpha z = 5$
is a line in R^3, then $\alpha + \beta$ is equal to : **[JEE(Main)- 20]**
(1) 10 (2) –10 (3) 2 (4) 0 **TD0102**

22. Let L be the line of intersection of planes $\vec{r}.(\hat{i}-\hat{j}+2\hat{k})=2$ and $\vec{r}.(2\hat{i}+\hat{j}-\hat{k})=2$. If $P(\alpha, \beta, \gamma)$ is the foot of perpendicular on L from the point (1,2,0), then the value of $35(\alpha+\beta+\gamma)$ is equal to :
[JEE(Main)- 2021]
(1) 101 (2) 119 (3) 143 (4) 134
TD0171

23. Let S be the mirror image of the point Q(1, 3, 4) with respect to the plane $2x - y + z + 3 = 0$ and let R(3, 5, γ) be a point of this plane. Then the square of the length of the line segment SR is _____.
[JEE(Main)- 2021]
TD0172

24. A line 'ℓ' passing through origin is perpendicular to the lines
$\ell_1 : \vec{r} = (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}$
$\ell_2 : \vec{r} = (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}$
If the co-ordinates of the point in the first octant on 'ℓ_2' at a distance of $\sqrt{17}$ from the point of intersection of 'ℓ' and 'ℓ_1' are (a, b, c), then $18(a + b + c)$ is equal to _____. **[JEE(Main)- 2021]**
TD0173

25. Let the image of the point P(1, 2, 3) in the line $L : \frac{x-6}{3} = \frac{y-1}{2} = \frac{z-2}{3}$ be Q. Let $R(\alpha, \beta, \gamma)$ be a point that divides internally the line segment PQ in the ratio 1 : 3. Then the value of $22(\alpha+\beta+\gamma)$ is equal to **[JEE(Main)- 2022]**
TD0174

26. If two distinct point Q, R lie on the line of intersection of the planes $-x + 2y - z = 0$ and $3x - 5y + 2z = 0$ and $PQ = PR = \sqrt{18}$ where the point P is (1, –2, 3), then the area of the triangle PQR is equal to
[JEE(Main)- 2022]
(A) $\frac{2}{3}\sqrt{38}$ (B) $\frac{4}{3}\sqrt{38}$ (C) $\frac{8}{3}\sqrt{38}$ (D) $\sqrt{\frac{152}{3}}$
TD0175

27. If the length of the perpendicular drawn from the point P(a, 4, 2), $a > 0$ on the line $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1}$ is $2\sqrt{6}$ units and $Q(\alpha_1, \alpha_2, \alpha_3)$ is the image of the point P in this line, then $a + \sum_{i=1}^{3} \alpha_i$ is equal to :
[JEE(Main)- 2022]
(A) 7 (B) 8 (C) 12 (D) 14
TD0176

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EXERCISE (JA)

1. Perpendiculars are drawn from points on the line $\frac{x+2}{2}=\frac{y+1}{-1}=\frac{z}{3}$ to the plane $x+y+z=3$. The feet of perpendiculars lie on the line **[JEE-Advanced 2013, 2]**

(A) $\frac{x}{5}=\frac{y-1}{8}=\frac{z-2}{-13}$ (B) $\frac{x}{2}=\frac{y-1}{3}=\frac{z-2}{-5}$ (C) $\frac{x}{4}=\frac{y-1}{3}=\frac{z-2}{-7}$ (D) $\frac{x}{2}=\frac{y-1}{-7}=\frac{z-2}{5}$

TD0113

2. A line ℓ passing through the origin is perpendicular to the lines

$\ell_1:(3+t)\hat{i}+(-1+2t)\hat{j}+(4+2t)\hat{k},-\infty<t<\infty$

$\ell_2:(3+2s)\hat{i}+(3+2s)\hat{j}+(2+s)\hat{k},-\infty<s<\infty$

Then , the coordinate(s) of the point(s) on ℓ_2 at a distance of $\sqrt{17}$ from the point of intersection of ℓ and ℓ_1 is(are) - **[JEE-Advanced 2013, 4, (–1)]**

(A) $\left(\frac{7}{3},\frac{7}{3},\frac{5}{3}\right)$ (B) (–1,–1,0) (C) (1,1,1) (D) $\left(\frac{7}{9},\frac{7}{9},\frac{8}{9}\right)$

TD0114

3. Two lines $L_1:x=5,\frac{y}{3-\alpha}=\frac{z}{-2}$ and $L_2:x=\alpha,\frac{y}{-1}=\frac{z}{2-\alpha}$ are coplanar. Then α can take value(s) **[JEE-Advanced 2013, 3, (–1)]**

(A) 1 (B) 2 (C) 3 (D) 4

TD0115

4. Consider the lines $L_1:\frac{x-1}{2}=\frac{y}{-1}=\frac{z+3}{1},L_2:\frac{x-4}{1}=\frac{y+3}{1}=\frac{z+3}{2}$ and the planes P_1:7x+y + 2z = 3, P_2 : 3x + 5y – 6z = 4. Let ax + by + cz = d be the equation of the plane passing through the point of intersection of lines L_1 and L_2 and perpendicular to planes P_1 and P_2.

Match List-I with List-II and select the correct answer using the code given below the lists.

List-I		List-II
P.	a =	1. 13
Q.	b =	2. –3
R.	c =	3. 1
S.	d =	4. –2

Codes :

	P	Q	R	S
(A)	3	2	4	1
(B)	1	3	4	2
(C)	3	2	1	4
(D)	2	4	1	3

[JEE-Advanced 2013, 3, (–1)]

TD0116

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Maths\Module\Vol-Module\Matrices, Vector & 3D\Eng\08-3D_(Exe.).p65

5. From a point P(λ,λ,λ), perpendiculars PQ and PR are drawn respectively on the lines y = x, z = 1 and y = –x, z = –1. If P is such that ∠QPR is a right angle, then the possible value(s) of λ is(are)

(A) $\sqrt{2}$ (B) 1 (C) –1 (D) $-\sqrt{2}$

[JEE(Advanced)-2014, 3]

TD0117

6. In $\mathbb{R}^3$, consider the planes $P_1 : y = 0$ and $P_2 : x + z = 1$. Let P_3 be a plane, different from P_1 and P_2, which passes through the intersection of P_1 and P_2. If the distance of the point (0,1,0) from P_3 is 1 and the distance of a point (α,β,γ) from P_3 is 2, then which of the following relations is (are) true ?

[JEE 2015, 4M, –2M]

(A) $2\alpha + \beta + 2\gamma + 2 = 0$ (B) $2\alpha - \beta + 2\gamma + 4 = 0$

(C) $2\alpha + \beta - 2\gamma - 10 = 0$ (D) $2\alpha - \beta + 2\gamma - 8 = 0$

TD0118

7. In $\mathbb{R}^3$, let L be a straight line passing through the origin. Suppose that all the points on L are at a constant distance from the two planes $P_1 : x + 2y - z + 1 = 0$ and $P_2 : 2x - y + z - 1 = 0$. Let M be the locus of the feet of the perpendiculars drawn from the points on L to the plane P_1. Which of the following points lie(s) on M ? **[JEE 2015, 4M, –2M]**

(A) $\left(0, -\frac{5}{6}, -\frac{2}{3}\right)$ (B) $\left(-\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right)$ (C) $\left(-\frac{5}{6}, 0, \frac{1}{6}\right)$ (D) $\left(-\frac{1}{3}, 0, \frac{2}{3}\right)$

TD0119

8. Consider a pyramid OPQRS located in the first octant ($x \geq 0, y \geq 0, z \geq 0$) with O as origin, and OP and OR along the x-axis and the y-axis, respectively. The base OPQR of the pyramid is a square with OP = 3. The point S is directly above the mid-point T of diagonal OQ such that TS = 3. Then-

(A) the acute angle between OQ and OS is $\frac{\pi}{3}$. **[JEE(Advanced)-2016, 4(–2)]**

(B) the equaiton of the plane containing the triangle OQS is x – y = 0

(C) the length of the perpendicular from P to the plane containing the triangle OQS is $\frac{3}{\sqrt{2}}$

(D) the perpendicular distance from O to the straight line containing RS is $\sqrt{\frac{15}{2}}$

TD0120

9. Let P be the image of the point (3, 1, 7) with respect to the plane x – y + z = 3. Then the equation of the plane passing through P and containing the straight line $\frac{x}{1} = \frac{y}{2} = \frac{z}{1}$ is

[JEE(Adv.)-2016, 3(–1)]

(A) x + y – 3z = 0 (B) 3x + z = 0 (C) x – 4y + 7z = 0 (D) 2x – y = 0

TD0121

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10. The equation of the plane passing through the point (1,1,1) and perpendicular to the planes $2x + y - 2z = 5$ and $3x - 6y - 2z = 7$, is- **[JEE(Advanced)-2017]**

(A) $14x + 2y + 15z = 31$ (B) $14x + 2y - 15z = 1$

(C) $-14x + 2y + 15z = 3$ (D) $14x - 2y + 15z = 27$

TD0122

11. Let $P_1 : 2x + y - z = 3$ and $P_2 : x + 2y + z = 2$ be two planes. Then, which of the following statement(s) is (are) TRUE ? **[JEE(Advanced)-2018, 4(–2)]**

(A) The line of intersection of P_1 and P_2 has direction ratios 1, 2, –1

(B) The line $\frac{3x-4}{9} = \frac{1-3y}{9} = \frac{z}{3}$ is perpendicular to the line of intersection of P_1 and P_2

(C) The acute angle between P_1 and P_2 is 60°

(D) If P_3 is the plane passing through the point (4, 2, –2) and perpendicular to the line of intersection of P_1 and P_2, then the distance of the point (2, 1, 1) from the plane P_3 is $\frac{2}{\sqrt{3}}$

TD0123

12. Let P be a point in the first octant, whose image Q in the plane x + y = 3 (that is, the line segment PQ is perpendicular to the plane x + y = 3 and the mid-point of PQ lies in the plane x + y = 3) lies on the z-axis. Let the distance of P from the x-axis be 5. If R is the image of P in the xy-plane, then the length of PR is _____ . **[JEE(Advanced)-2018, 3(0)]**

TD0124

13. Consider the cube in the first octant with sides OP, OQ and OR of length 1, along the x-axis, y-axis and z-axis, respectively, where O(0, 0, 0) is the origin. Let $S\left(\frac{1}{2},\frac{1}{2},\frac{1}{2}\right)$ be the centre of the cube and T be the vertex of the cube opposite to the origin O such that S lies on the diagonal OT. If $\vec{p} = \overrightarrow{SP}$, $\vec{q} = \overrightarrow{SQ}$, $\vec{r} = \overrightarrow{SR}$ and $\vec{t} = \overrightarrow{ST}$, then the value of $\left|(\vec{p}\times\vec{q})\times(\vec{r}\times\vec{t})\right|$ is _____ .

[JEE(Advanced)-2018, 3(0)]

TD0125

14. Let L_1 and L_2 denotes the lines

$$\vec{r} = \hat{i} + \lambda(-\hat{i} + 2\hat{j} + 2\hat{k}), \lambda \in \mathbb{R}$$

and $$\vec{r} = \mu(2\hat{i} - \hat{j} + 2\hat{k}), \mu \in \mathbb{R}$$

respectively. If L_3 is a line which is perpendicular to both L_1 and L_2 and cuts both of them, then which of the following options describe(s) L_3 ? **[JEE(Advanced)-2019, 4(–1)]**

(1) $\vec{r} = \frac{1}{3}(2\hat{i} + \hat{k}) + t(2\hat{i} + 2\hat{j} - \hat{k}), t \in \mathbb{R}$ (2) $\vec{r} = \frac{2}{9}(2\hat{i} - \hat{j} + 2\hat{k}) + t(2\hat{i} + 2\hat{j} - \hat{k}), t \in \mathbb{R}$

(3) $\vec{r} = t(2\hat{i} + 2\hat{j} - \hat{k}), t \in \mathbb{R}$ (4) $\vec{r} = \frac{2}{9}(4\hat{i} + \hat{j} + \hat{k}) + t(2\hat{i} + 2\hat{j} - \hat{k}), t \in \mathbb{R}$

TD0126

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15. Three lines are given by

$\vec{r} = \lambda\hat{i},\ \lambda \in \mathbb{R}$

$\vec{r} = \mu(\hat{i} + \hat{j}),\ \mu \in \mathbb{R}$ and

$\vec{r} = \nu(\hat{i} + \hat{j} + \hat{k}),\ \nu \in \mathbb{R}$.

Let the lines cut the plane x + y + z = 1 at the points A, B and C respectively. If the area of the triangle ABC is Δ then the value of $(6\Delta)^2$ equals ___ **[JEE(Advanced)-2019, 3(0)]**

TD0127

16. Let L_1 and L_2 be the following straight lines. **TD0149**

$L_1 : \frac{x-1}{1} = \frac{y}{-1} = \frac{z-1}{3}$ and $L_2 : \frac{x-1}{-3} = \frac{y}{-1} = \frac{z-1}{1}$

Suppose the straight line

$L : \frac{x-\alpha}{l} = \frac{y-1}{m} = \frac{z-\gamma}{-2}$ **[JEE(Advanced)-2020]**

lies in the plane containing L_1 and L_2, and passes through the point of intersection of L_1 and L_2. If the line L bisects the acute angle between the lines L_1 and L_2, then which of the following statements is/are TRUE?

(A) $\alpha - \gamma = 3$ (B) $\ell + m = 2$ (C) $\alpha - \gamma = 1$ (D) $\ell + m = 0$

17. Let $\alpha, \beta, \gamma, \delta$ be real numbers such that $\alpha^2 + \beta^2 + \gamma^2 \neq 0$ and $\alpha + \gamma = 1$. Suppose the point (3,2,–1) is the mirror image of the point (1,0,–1) with respect to the plane $\alpha x + \beta y + \gamma z = \delta$. Then which of the following statements is/are TRUE ? **[JEE(Advanced)-2020]**

(A) $\alpha + \beta = 2$ (B) $\delta - \gamma = 3$ (C) $\delta + \beta = 4$ (D) $\alpha + \beta + \gamma = \delta$

TD0150

Question Stem for Question Nos. 18 and 19

Question Stem

Let α, β and γ be real numbers such that the system of linear equations

$x + 2y + 3z = \alpha$

$4x + 5y + 6z = \beta$

$7x + 8y + 9z = \gamma - 1$

is consistent. Let |M| represent the determinant of the matrix $M = \begin{bmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

Let P be the plane containing all those (α, β, γ) for which the above system of linear equations is consistent, and D be the square of the distance of the point (0, 1, 0) from the plane P.

18. The value of |M| is _____. **TD0151**

19. The value of D is _____. **TD0152**

[JEE(Advanced)-2021]

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20. Let P_1 and P_2 be two planes given by

$P_1 : 10x + 15y + 12z - 60 = 0,$

$P_2 : -2x + 5y + 4z - 20 = 0.$

Which of the following straight lines can be an edge of some tetrahedron whose two faces lie on P_1 and P_2 ? **[JEE(Advanced)-2022]**

(A) $\frac{x-1}{0} = \frac{y-1}{0} = \frac{z-1}{5}$ (B) $\frac{x-6}{-5} = \frac{y}{2} = \frac{z}{3}$

(C) $\frac{x}{-2} = \frac{y-4}{5} = \frac{z}{4}$ (D) $\frac{x}{1} = \frac{y-4}{-2} = \frac{z}{3}$ **TD0177**

21. Let S be the reflection of a point Q with respect to the plane given by

$$\vec{r} = -(t+p)\hat{i} + t\hat{j} + (1+p)\hat{k}$$

where t, p are real parameters and $\hat{i}, \hat{j}, \hat{k}$ are the unit vectors along the three positive coordinate axes.

If the position vectors of Q and S are $10\hat{i} + 15\hat{j} + 20\hat{k}$ and $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ respectively, then which of the following is/are TRUE ? **[JEE(Advanced)-2022]**

(A) $3(\alpha + \beta) = -101$ (B) $3(\beta + \gamma) = -71$

(C) $3(\gamma + \alpha) = -86$ (D) $3(\alpha + \beta + \gamma) = -121$ **TD0178**

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ANSWER KEY

Do yourself 1

1. $2\sqrt{21}$ **3.** $8x + 2y + 24z \pm 2k^2 + 9 = 0$ **4.** $\left(-2, \frac{8}{3}, 5\right)$ & $\left(-1, \frac{10}{3}, 6\right)$

5. **(a)** 7 : 8, externally **(b)** 2 : 3 internally

Do yourself - 2

1. 3, 2, 5 **2.** 3, 4, 3

3. 2,3,4 & $\frac{2}{\sqrt{29}}, \frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}}$ or $-\frac{2}{\sqrt{29}}, -\frac{3}{\sqrt{29}}, -\frac{4}{\sqrt{29}}$

Do yourself - 3

1. $\theta = \frac{\pi}{2}$ **3.** $\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}$ or $-\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}$ **4.** $\theta = 90°$

5. C **6.** A,D

Do yourself - 4

1. **(a)** $\frac{3}{5\sqrt{2}}, \frac{4}{5\sqrt{2}}, \frac{1}{\sqrt{2}}$ **(b)** $\left(\frac{7}{3}, 0, 0\right), \left(0, \frac{7}{4}, 0\right)$ & $\left(0, 0, \frac{7}{5}\right)$ **(c)** $\frac{x}{7/3} + \frac{y}{7/4} + \frac{z}{7/5} = 1$

(d) $\frac{3x}{5\sqrt{2}} + \frac{4y}{5\sqrt{2}} + \frac{z}{\sqrt{2}} = \frac{7}{5\sqrt{2}}$ & $\vec{r}.\left(\frac{3}{5\sqrt{2}}\hat{i} + \frac{4}{5\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}\right) = \frac{7}{5\sqrt{2}}$

2. $x + y + 2z = 7$ **3.** B **4.** D

Do yourself - 5

2. $\theta = \cos^{-1}\left(\frac{1}{\sqrt{156}}\right)$

Do yourself - 6

1. $\frac{8}{\sqrt{6}}$ **2.** $3x + 4y + 2z = 0$ **3.** A

Do yourself - 7

1. P, Q same side & R opposite side

2. **(a)** $4x + y - 5z + 14 = 0$ & $32x - 13y + 23z - 56 = 0$

(b) $4x + y - 5z - 14 = 0$ & $32x - 13y + 23z - 70 = 0$

(c) $4x + y - 5z + 14 = 0$ (obtuse angle bisector) & $32x - 13y + 23z - 56 = 0$ (acute angle bisector)

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Do yourself - 8

1. $\frac{x-4}{1}=\frac{y-2}{-1}=\frac{z-3}{2}$ **2.** $\frac{x-11/4}{-3}=\frac{y+9/4}{5}=\frac{z-0}{4}$ **3.** $\theta=\sin^{-1}\left(\frac{6}{5\sqrt{5}}\right)$

5. B **6.** $\frac{x-1}{2}=\frac{y-2}{2}=\frac{z-3}{-3}$ **7.** B

Do yourself - 9

1. $\left(\frac{-5}{3},\frac{13}{3},\frac{2}{3}\right)$ & $\left(\frac{-1}{3},\frac{11}{3},\frac{4}{3}\right)$ **2.** 1 **4.** $\frac{x}{2}+\frac{y}{3}+\frac{z}{-5}=1$, Area $=\frac{19}{2}$ sq. units

5. C **6.** A **7.** A **8.** B,C

Do yourself - 10

1. $\left(\frac{1}{2},-\frac{1}{2},-\frac{3}{2}\right)$ **2.** $\frac{\sqrt{1218}}{2}$ **3.** C

4. $\frac{x-7}{22}=\frac{y-2}{5}=\frac{z+1}{-4}$ **5.** D **6.** C

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	D	A	B	D	B	A	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	C	C	D	A	C	B	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	C	B	C	A	B	A	A	B

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	A,D	A,D	A,B,C	A,C	A,C	A,B,D	A,D	C,D	B,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,C	B,C,D	B,D	C,D	B,C,D	A,B,C,D	B,C	B,C	B	A,B

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EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	C	D	C	A	A,B,C	A,B,C	1.41 or 1.42	0.66 or 0.67	D	
Q.10	A	B	C	D						
	R	Q	P	S						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	17	6	9	6	3	4.50	3.50	4.00	2.44 or 2.45

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	1	2	3	1	1	1	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	3	1	2	2	2	1	4	2
Que.	21	22	23	24	25	26	27			
Ans.	1	2	72	44	125	B	B			

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B,D	A,D	A	C	B,D	A,B	B,C,D	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C,D	8	0.50	1,2,4	0.75	A,B	A,B,C	1.00	1.50	A,B,D
Que.	21									
Ans.	A,B,C									

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